

Apala Pramanik

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PERSONAL PROFILE

PhD researcher at UNL specializing in ML-driven beam management and power control for mmWave 5G/6G vehicular networks (V2I/V2X). Experienced in SDR-based hardware-in-the-loop testbeds (USRP/GNU Radio), ray-tracing channel modeling (Wireless InSite, Sionna RT), with expertise in deep reinforcement learning for wireless PHY-layer optimization. Currently expanding into O-RAN architectures.

EDUCATION

University of Nebraska–Lincoln – PhD, Computer Science

GPA: 3.75/4.0 | Advisor: Dr. Mehmet Can Vuran, Dale M. Jensen Chair Professor

Dissertation: ML-Driven Beam Management and Power Control for mmWave Vehicular Networks

June 2024 – Present

Nebraska, USA

University of Nebraska–Lincoln – Master's, Computer Science

GPA: 3.75/4.0 | Advisor: Dr. Dung Hoang Tran, Assistant Professor

Aug 2021 – May 2024

Nebraska, USA

Guru Gobind Singh Indraprastha University – B.Tech, Electronics and Communication

GPA: 7.9/10 | Thesis: Deep learning techniques for MRI denoising and reconstruction from under sampled MRI scans

Delhi, India

RESEARCH EXPERIENCE

University of Nebraska–Lincoln

Graduate Research Assistant – PhD

Nebraska, USA

Aug 2024 – Present

- Camera-Primed Double-Directional Beam Management (IEEE SECON 2026): Developed a vision-assisted mmWave beamforming framework enabling real-time TX/RX beam selection using camera-derived spatial cues and adaptive correction under hardware constraints.
- Reinforcement Learning for mmWave Power Control: Designed a hardware-in-the-loop RL framework to optimize transmit power and RF gain parameters under SNR constraints using USRP/GNU Radio testbeds and real-time PHY-layer feedback.
- Utilized ray-tracing tools (Wireless InSite, Sionna RT) to model mmWave propagation environments and validate experimental channel observations.
- Technical Skills:** GNU Radio, USRP, Python, C++, MATLAB, CUDA/PyTorch, Ubuntu Linux, Git

Graduate Research Assistant – Master's

Aug 2021 – May 2024

- Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems (ACM/IEEE MEMOCODE 2024).
- Applied Neural Network Verification (NNV) for formal analysis of deep learning models in safety-critical systems.
- Specified and verified temporal properties in cyber-physical systems using Signal Temporal Logic (STL).
- Technical Skills:** ROS (rospy, roscpp), Gazebo, RVIZ, Python, C++, Ubuntu Linux, Git

PUBLICATIONS

Conference Proceedings

Look Once, Beam Twice: Camera-Primed Real-Time Double-Directional mmWave Beam Management for Vehicular Connectivity

Apala Pramanik, Avhishek Biswas, Eylem Ekici, Mehmet Can Vuran

IEEE International Conference on Sensing, Communication, and Networking (SECON), 2026

Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems

Apala Pramanik, Sung Woo Choi, Yuntao Li, Luan Viet Nguyen, Kyungki Kim, Hoang-Dung Tran

22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE), 2024

ASTITVA: Assistive Special Tools and Technologies for Inclusion of Visually Challenged

Apala Pramanik, Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Rohan Tripathi

International Conference on Computing, Communication, and Intelligent Systems (ICCCIS), 2021

START: Smart Stick based on TLC Algorithm in IoT Network for Visually Challenged Persons

Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Apala Pramanik

Fourth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud), 2020

Workshops

Vision-based Runtime Monitoring for Human-Construction Robot Systems

Apala Pramanik, Kyungki Kim, Dung Hoang Tran

IROS 2023 Workshop on Formal Methods Techniques in Robotics Systems: Design and Control, 2023

UNIVERSITY PROJECTS

Multi-Modal Traffic Pattern Detection Using Radio and Camera Sensing

Fall 2024

- Built a real-time traffic classification system using synchronized USRP RF signals and camera images; real- and complex-valued CNNs achieved >80% accuracy with <100 ms inference latency via GPU-accelerated execution (CUDA/PyTorch).
- Integrated ± 10 ms timestamp synchronization and watchdog mechanism for reliable, continuous operation.
- **Technical Skills:** Python, PyTorch, CUDA, USRP, Signal Processing, Deep Learning (CNN, CVNN), Multi-Modal Data Fusion

mmWave 3D Ray Tracing Framework (MATLAB)

Fall 2024

- Built a MATLAB-based 3D ray tracing simulator for mmWave indoor propagation (24 GHz) with multi-bounce reflections, log-normal shadowing, and link budget computation.
- Developed modular environment modeling with power-gradient visualization and automated SVG/EPS/GIF export (Python).
- **Technical Skills:** MATLAB, Python, Ray Tracing, mmWave Channel Modeling, Link Budget Analysis

Deep Learning and NLP Projects

Aug 2021 – July 2024

- Developed a Spanish news classification system using GRU, LSTM, and BETO Transformer models, achieving 91.80% accuracy with tokenizer optimization and data augmentation.
- Implemented CNN image classifiers on Fashion-MNIST and CIFAR-100; sentiment analysis using RNN and Transformer architectures.
- **Technical Skills:** Python, PyTorch, TensorFlow, Pandas, Deep Learning (CNNs, RNNs, Transformers)

INVOLVEMENT

Peer Reviewer	IEEE Transactions on Wireless Communications (2025, 2026)
Undergrad Research Fair	Presented research to undergraduate Computer Science students at UNL
K–12 Outreach	Demonstrated research activities to K–12 students
Mentorship	Mentored undergraduate researchers on experimental design, data analysis, and research communication

SKILLS

ML / DL	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, CUDA
Wireless / SDR	GNU Radio, USRP, Wireless InSite, Sionna RT, mmWave, 5G/6G, V2X/V2I, Beamforming, OFDM
Languages	Python, C++, MATLAB, Bash/Zsh
Robotics / CPS	ROS (rospy, roscpp), Gazebo, RVIZ, Point Cloud Library
Tools	Git, Linux, LaTeX, Arduino, Proteus
Coursework	Deep Learning (A+), Wireless Comm Networks (A+), Advanced Wireless Comm (A), Hardware Acceleration for ML/CUDA (B+), IoT (A+)
Soft Skills	Mentoring, Time Management, Teamwork, Problem-solving, Documentation, Engaging Presentation

INTERESTS

Paper Review	Critically evaluating technical rigor, methodology, and scientific contributions in peer-reviewed research.
Tech Writing	Conveying complex research clearly and accessibly through papers and documentation.
Bharatnatyam	Senior diploma holder with 10+ years of training in Indian Classical Dance.
Fitness	Passionate about weight training and maintaining a healthy and active lifestyle.

References available upon request.